Elliptic-Curve Cryptography as a Quotient of Scalar History Space

Abstract. Elliptic-curve cryptographic systems derive security not merely from the hardness of the discrete logarithm problem, but from a deliberate erasure of scalar provenance. All observable values lie in a single cyclic subgroup generated by a base point G, and are obtained via a quotient map that collapses all scalar histories to indistinguishable group elements. This document formalizes that viewpoint using a conceptual history space whose projection admits no computable section. Security failures correspond precisely to protocol-level leaks that expose relations between multiple representatives of the same equivalence class.

1. Scalar Action and Quotient Structure

Let E be an elliptic curve over a finite field with a prime-order subgroup generated by G, of order n. Scalar multiplication defines a surjective homomorphism ϕ : Z_n $\to$ E, mapping a scalar d to the point dG. This map induces a quotient that erases all scalar provenance: any decomposition of d into keys, nonces, or intermediate values becomes indistinguishable once mapped into the group.

2. History Space Interpretation

We define a conceptual history space $\blacksquare$ whose elements encode both a curve point and the scalar relations that produced it, such as nonce usage, linear combinations, or transaction context. The elliptic-curve group E is obtained as a quotient of this space via a projection $\pi : \blacksquare \to E$. All observable cryptographic values are images under π.

3. Non-Existence of a Section

Security relies on the fact that there exists no efficient section of either π or ϕ. Any computable lift from E back to scalar space would constitute a solution to the discrete logarithm problem. Thus, scalar provenance is intentionally and irreversibly destroyed by the group action.

4. Protocol Variables as Projections

In ECDSA, protocol variables such as x, y, r, s, and h represent distinct projections of hidden scalar expressions. Each projection discards specific information, and no two projections align to form a solvable system unless the protocol invariant is violated.

5. Failure Modes

All known elliptic-curve signature failures correspond to violations of the quotient invariant, including nonce reuse, linear nonce relations, biased nonces, or side-channel leakage. Each failure effectively reintroduces a partial section of the quotient, enabling scalar recovery.

6. Conclusion

Elliptic-curve cryptography is best understood as operating on a quotient of a richer scalar history space. Its security arises from the intentional erasure of scalar provenance. Attacks succeed only when protocol errors expose relations between multiple representatives of the same equivalence class.

Core invariant: Elliptic-curve cryptography does not hide the axis; it destroys the coordinates.